

SYLTRA

Smart Living. Seamlessly Connected.

OCCUPATIONAL HEALTH & SAFETY

ISO 45001:2018

Health & Safety

ISO 45001:2018

Electrical safety, working-at-height & emergency response

OH&S policy, hazard procedures, risk matrix & work permit

1. OH&S Context & Scope

SYLTRA's field operations — live electrical panel work, PLC/SCADA commissioning, and cable-tray installation — carry inherent occupational hazards. This scope covers the identification, control, and continual improvement of health and safety performance across every site engagement.

2. Occupational Health & Safety Policy

As part of its Integrated Management System, SYLTRA's management commits to:

- Providing a safe and healthy work environment that prevents injuries and occupational illness — specifically electrical-shock hazards and mechanical injuries during on-site installation.
- Complying with all applicable occupational safety legislation and regulatory requirements.
- Continually improving safety performance and involving employees in decisions related to workplace safety.

3. Procedure — Hazard Identification & OH&S Risk Management

Objective: Identify hazards associated with automation installation and maintenance operations, assess them, and establish controls.

- Before any field project begins, the site engineer completes a Job Hazard Analysis (JHA) form.
- Hazards are classified based on (Probability × Severity) against the approved risk matrix.
- The hierarchy of controls is applied: elimination, substitution, engineering controls (e.g. electrical isolation / LOTO), administrative controls, and Personal Protective Equipment (PPE).
- The register is updated annually or whenever new technologies or equipment are introduced.

4. Procedure — Emergency Preparedness & Response

Objective: Establish an action plan for electrical-shock emergencies, control-panel fires, or field incidents.

- Equip every work site with CO2-class fire extinguishers designated for electrical fires, plus first-aid kits.
- Train 100% of technicians in electrical-shock rescue maneuvers and basic first aid.
- Immediately cut power at the main disconnect (Emergency Switch) when any emergency occurs, before attending to the casualty.

5. HSE Risk Assessment Matrix (Safety Extract)

Process / Activity	Potential Hazard	Health Impact	P	S	Level	Required Controls
On-site installation of automation panels	Electrical shock during wiring and voltage testing	Serious injury or fatality	2	5	10 (Med)	Apply LOTO isolation, use fully insulated tools, and wear electrically insulated safety footwear.
Cable routing at elevated locations	Fall from height	Fractures and serious lacerations	3	4	12 (High)	Use certified scaffolding; wear a full-body fall-arrest harness anchored to a secure point.

6. Template — Electrical Work Permit

Permit Date: _____ Site / Project Name: _____

Detailed Work Description: _____

Pre-Start Checklist

- Main electrical current disconnected and Lock-Out Tag-Out (LOTO) applied.
- Absence of voltage verified using a certified test screwdriver and multimeter.
- All technicians wearing PPE (helmet, goggles, insulated footwear, high-voltage gloves).
- CO2 fire extinguisher available and ready in the immediate work area.

Safety / Site Engineer Approval: _____ Signature: _____

7. Relevant Internal Audit Checkpoints (ISO 45001:2018)

Clause	Audit Requirement	Verification Method & Evidence	Conforms?
Clause 5: Leadership	Does management commit to funding insulated safety tools and training engineers on safe automation practice?	Review top-management meeting minutes and PPE/original-software purchase budget.	
Clause 6: Planning	Have safety objectives been defined (e.g. zero electrical incidents)?	Inspect the "Quality, Safety & Environmental Objectives" document for the current year.	
Clause 7: Support	Are engineers and technicians qualified with certifications for handling production lines and electrical voltage?	Review HR files, PLC training certificates, and electrical-safety training records.	
Clause 8: Operation	How are sudden on-site programming changes controlled to protect machine safety and product quality?	Inspect Change Management Records and customer sign-offs.	
Clause 10: Improvement	Is a corrective-action report (NCR) opened when a programming fault or mechanical failure occurs during installation?	Inspect the corrective-action log and verify root-cause methods (5 Whys) are applied.	